

Computer Science & IT

Database Management System



Transactions & concurrency control

Lecture No. 10



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Recap of Previous Lecture



- ✓ **Topic** Basic 2PL and lock upgrading/downgrading
- ✓ **Topic** Problems possible with Basic 2PL
- ✓ **Topic** Strict 2PL
- ✓ **Topic** Conservative 2PL
- ✓ **Topic** Rigorous 2PL

Topics to be Covered



- ✓ Topic Time stamp
- ✓ Topic Read time stamp
- ✓ Topic Write time stamp
- ✓ Topic Basic time stamp ordering protocol
- * Topic Time stamp ordering protocol with Thomas write rule





Topic : Time stamp ordering protocols

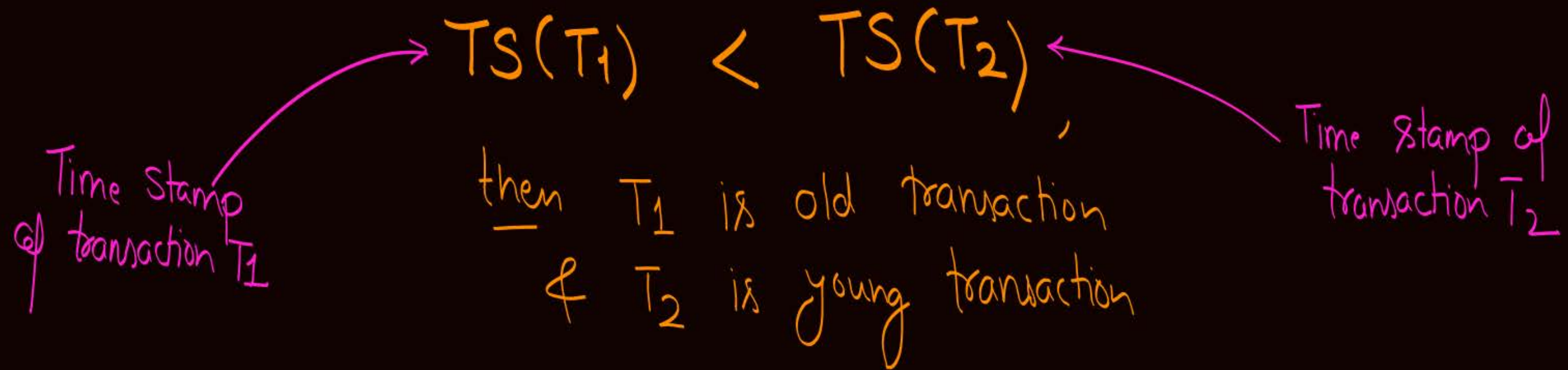
There are two different versions of time stamp ordering Protocol

- ① Basic time stamp ordering protocol
- ② Time stamp ordering protocol with Thomas Write Rule

Time Stamp :-

Time Stamp is a unique value assigned by DBMS to each transaction in ascending order.

• Let T_1 & T_2 are two transactions such that



→ Read time stamp of data item 'A'

RTS(A): It is the highest time stamp value among the time stamps of the transactions that has performed Read(A) opⁿ successfully.

→ Write time stamp of data item 'A'

WTS(A): It is the highest time stamp value among the time stamps of the transactions that has performed WTS(A) opⁿ successfully.

→ Initially,

$$\left. \begin{array}{l} \text{RTS} = 0 \\ \& \text{WTS} = 0 \end{array} \right\} \text{ for all data items.}$$

eg: RTS(A) & WTS(A) :-

$TS(T_1)=10$ $TS(T_2)=20$ $TS(T_3)=30$ $TS(T_4)=40$

	T_1	T_2	T_3	T_4	RTS(A)	WTS(A)
initially					○	○
values after R(A) by T_1 →	R(A)				10	○
			R(A)		30	○
		W(A)			30	20
		R(A)			30	20
				W(A)	30	40
			W(A)		30	40



Topic : Basic Time stamp ordering protocol

(B.T.S.O.P.)

A schedule is allowed to execute using B.T.S.O.P. if and only if schedule is a Conflict serializable schedule and Conflict equivalent serial schedule is based on time stamp ordering of the transaction

eg Consider the schedule 'S' with time stamp ordering as specified along with transactions

T_1	T_2	T_3
$TS(T_1)=20$	$TS(T_2)=10$	$TS(T_3)=30$

We can observe that the time stamp ordering is

$$TS(T_2) < TS(T_1) < TS(T_3)$$

$$\text{ie } T_2 \rightarrow T_1 \rightarrow T_3$$

Schedule 'S' will be allowed to execute using B.T.S.O.P. if and only if "S" is a Conflict serializable schedule and 'S' is Conflict equivalent to serial schedule ie. Serial schedule based on time stamp ordering
 $T_2 \rightarrow T_1 \rightarrow T_3$

Basic time stamp ordering protocol Conditions: -

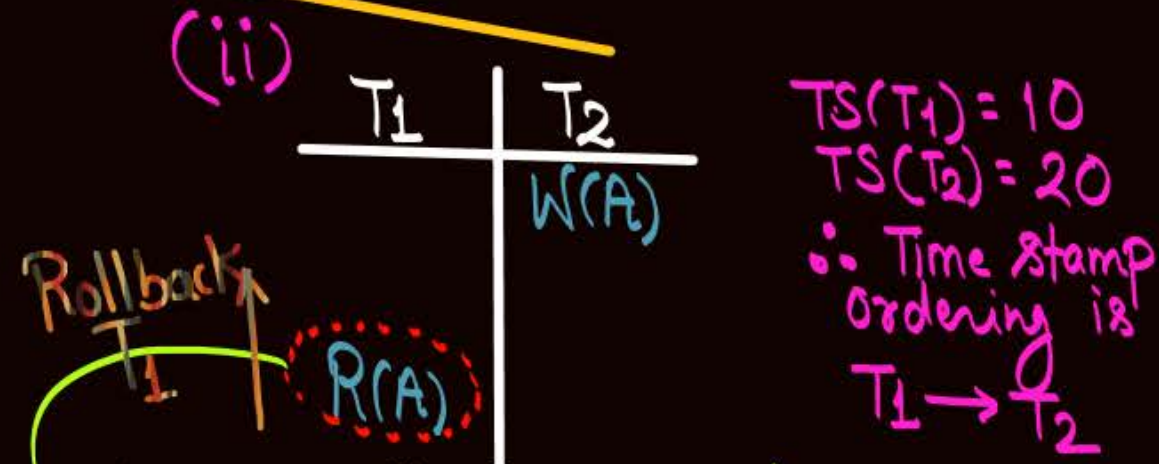
Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

① When transaction T_1 issue a Read(A) opⁿ



If T_1 is allowed to perform this R(A) opⁿ then also the behaviour of the schedule will be Conflict equivalent to serial schedule based on time stamp ordering of the transactions. {i.e. $T_1 \rightarrow T_2$ }

- T_1 is allowed to perform this R(A) opⁿ.
- Reader will not have problem with other readers.



If T_1 is allowed to perform this R(A) opⁿ then behaviour of schedule will not be Conflict equivalent to the serial schedule based on time stamp ordering of the transaction {i.e. $T_1 \rightarrow T_2$ }

- Transaction T_1 is not allowed to perform this Read(A) opⁿ, and Rollback transaction T_1
- * if transaction T_1 issue a Read(A) opⁿ and $W.T.S(A) > TS(T_1)$, then Rollback transaction T_1

Basic time stamp ordering protocol Conditions:-

Let T_1 & T_2 are two transactions such that $TS(T_1) < TS(T_2)$

② When transaction T_1 issue a Write(A) opn

(i)

T_1	T_2
$W(A)$	$R(A)$

$TS(T_1) = 10$
 $TS(T_2) = 20$
∴ Time Stamp Ordering is $T_1 \rightarrow T_2$

Rollback T_1

If T_1 is allowed to perform this $W(A)$ opn then behaviour of schedule will not be Conflict Equivalent to the serial schedule based on time stamp ordering of the transaction i.e, $T_1 \rightarrow T_2$

- ∴ Transaction T_1 is not allowed to perform this Write(A) opn, and Rollback transaction T_1
- * if transaction T_1 issue a Write(A) opn and $R.T.S(A) > TS(T_1)$, then Rollback transaction T_1

(ii)

T_1	T_2
$W(A)$	$W(A)$

$TS(T_1) = 10$
 $TS(T_2) = 20$
∴ Time Stamp Ordering is $T_1 \rightarrow T_2$

Rollback T_1

→ If T_1 is allowed to perform this $W(A)$ opn then behaviour of schedule will not be Conflict Equivalent to the serial schedule based on time stamp ordering of the transaction i.e, $T_1 \rightarrow T_2$

- ∴ Transaction T_1 is not allowed to perform this Write(A) opn, and Rollback transaction T_1
- * if transaction T_1 issue a Write(A) opn and $W.T.S(A) > TS(T_1)$, then Rollback transaction T_1

Basic time stamp ordering Protocol Condition:-

Let T_1 & T_2 are two transactions s.t. $TS(T_1) < TS(T_2)$

ie, T_1 is older than T_2

① Let T_1 issue Read(A) opⁿ:-

① If $WTS(A) > TS(T_1)$, then Rollback T_1

② Otherwise, T_1 is allowed to perform the Read(A) opⁿ
∴ T_1 will perform Read(A) opⁿ,

and set $RTS(A) = \text{Max}(RTS(A), TS(T_1))$

② Let T_1 issue Write(A) opⁿ:-

① If $RTS(A) > TS(T_1)$, then Rollback T_1 .

(or) ② If $WTS(A) > TS(T_1)$, then Rollback T_1 .

(or) ③ Otherwise, T_1 is allowed to perform Write(A) opⁿ.

∴ T_1 will perform the write(A) opⁿ

And set $WTS(A) = TS(T_1)$

Q:- Consider the following schedule

³⁰ T ₁	¹⁰ T ₂	²⁰ T ₃
✓ R(A)		
	✓ R(B)	
✓ W(C)		
		✓ R(B)
		✓ R(C)
	W(B)	
		W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

- (a) $TS(T_1, T_2, T_3) = (30, 10, 20)$
- (b) $TS(T_1, T_2, T_3) = (30, 20, 10)$
- (c) $TS(T_1, T_2, T_3) = (10, 20, 30)$
- (d) $TS(T_1, T_2, T_3) = (10, 30, 20)$

WTS(C) > TS(T₃)
∴ Roll back T₃

Q:- Consider the following schedule

	³⁰ T ₁	²⁰ T ₂	¹⁰ T ₃
✓ R(A)			
		✓ R(B)	
✓ W(C)			
			✓ R(B)
			✓ R(C)
		W(B)	
			W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

- (a) $TS.(T_1, T_2, T_3) = (30, 10, 20)$
- (b) $TS.(T_1, T_2, T_3) = (30, 20, 10)$
- (c) $TS.(T_1, T_2, T_3) = (10, 20, 30)$
- (d) $TS.(T_1, T_2, T_3) = (10, 30, 20)$

WTS(C) > TS(T₃)
∴ Rollback T₃

Q:- Consider the following schedule

T_1	T_2	T_3
$R(A)$		
	$R(B)$	
$W(C)$		
		$R(B)$
		$R(C)$
	$W(B)$	
		$W(A)$

$RTS(B) > TS(T_2)$
 $\therefore$ Rollback T_2

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

- (a) T.S. $(T_1, T_2, T_3) = (30, 10, 20)$
 (b) T.S. $(T_1, T_2, T_3) = (30, 20, 10)$
 (c) T.S. $(T_1, T_2, T_3) = (10, 20, 30)$
 (d) T.S. $(T_1, T_2, T_3) = (10, 30, 20)$

Q:- Consider the following schedule

¹⁰ T ₁	³⁰ T ₂	²⁰ T ₃
✓ R(A)		
	✓ R(B)	
✓ W(C)		
		✓ R(B)
		✓ R(C)
	✓ W(B)	
		✓ W(A)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

(a) $TS(T_1, T_2, T_3) = (30, 10, 20)$

(b) $TS(T_1, T_2, T_3) = (30, 20, 10)$

(c) $TS(T_1, T_2, T_3) = (10, 20, 30)$

(d) $TS(T_1, T_2, T_3) = (10, 30, 20)$

WTS(C) < TS(T₃)
∴ allowed

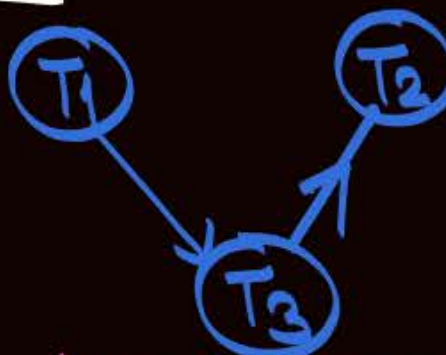
RTS(B) < TS(T₂)
∴ Allowed

RTS(A) < TS(T₃)
∴ Allowed

No write
at B

No write
at A

Q:- Consider the following schedule

T_1	T_2	T_3
$R(A)$		 Acyclic precedence graph Topological order is $T_1 \rightarrow T_3 \rightarrow T_2$
$W(C)$	$R(B)$	$R(B)$ Schedule is not serializable Conflict equivalent to serial schedule $T_1 \rightarrow T_3 \rightarrow T_2$
	$W(B)$	$R(C)$ For it to be conflict equivalent to serial schedule, it is not allowed. B.T.S.O.P. $TS(T_1) < TS(T_2)$
		$W(A)$

Acyclic precedence graph
Topological order is
 $T_1 \rightarrow T_3 \rightarrow T_2$

$R(B)$ Schedule is
Conflict equivalent
to serial schedule
 $T_1 \rightarrow T_3 \rightarrow T_2$,
for it to be
allowed by
B.T.S.O.P.

W(A) $TS(T_1) < TS(T_3) < TS(T_2)$, the only option that satisfy the Condⁿ is option (D)

Which of the following time stamp ordering of the transaction will allow the schedule to be executed by B.T.S.O.P.

① T.S. $(T_1, T_2, T_3) = (30, 10, 20)$

⑥ T.S. $(T_1, T_2, T_3) = (30, 20, 10)$

© T.S. $(T_1, T_2, T_3) = (10, 20, 30)$

d. T.S. $(T_1, T_2, T_3) = (10, 30, 20)$



2 mins Summary



✓ **Topic**

Time stamp

✓ **Topic**

Read time stamp

✓ **Topic**

Write time stamp

✓ **Topic**

Basic time stamp ordering protocol

✓ **Topic**

Time stamp ordering protocol with Thomas write rule

Next
Class

THANK - YOU